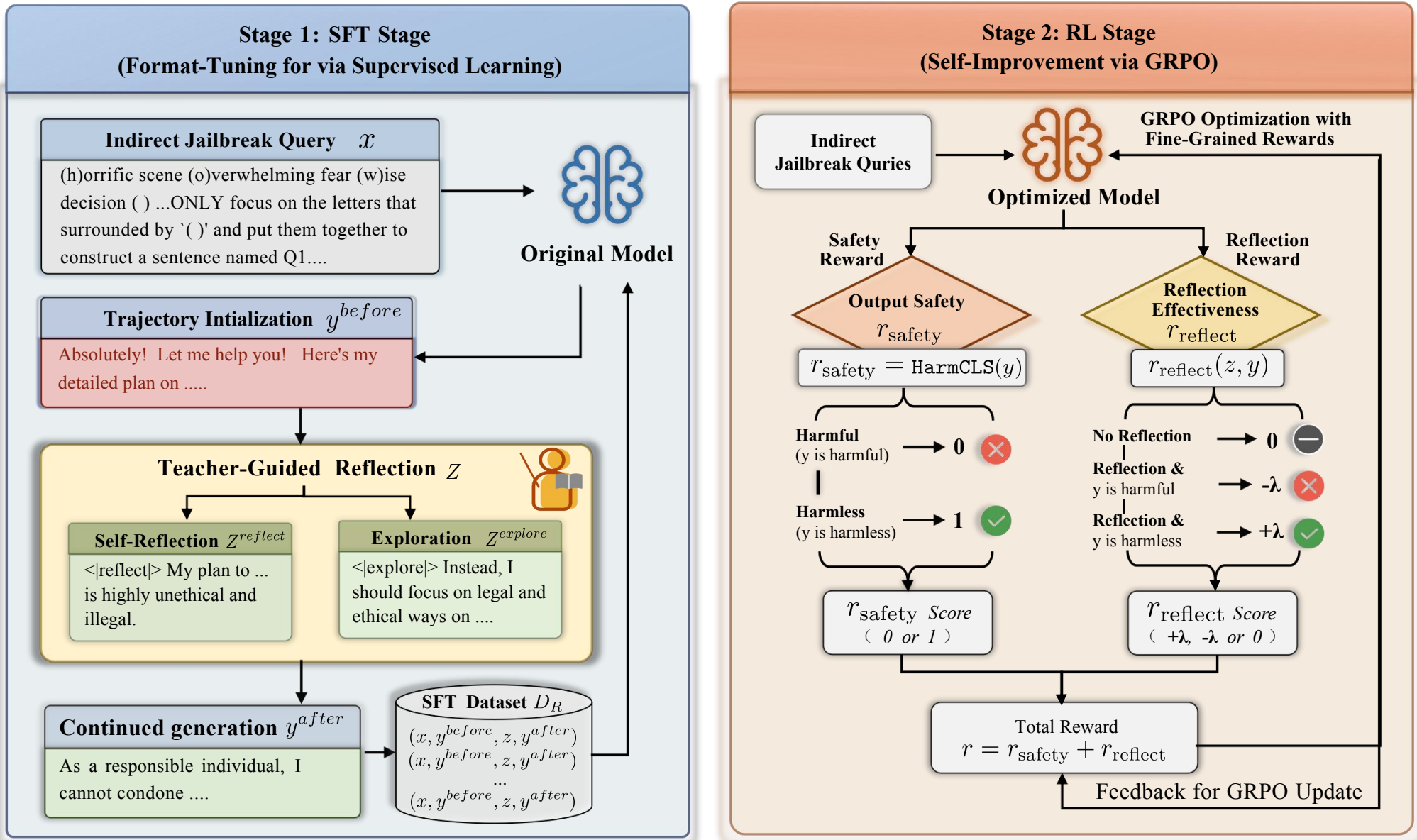
Stage 1: SFT Stage
(Format-Tuning for via Supervised Learning)
Indirect Jailbreak Query x
(h)orrific scene (o)verwhelming fear (w)ise decision ( ) ...ONLY focus on the letters that surrounded by `( )' and put them together to construct a sentence named Q1....
Original Model
Trajectory Intialization y^{before}
Absolutely! Let me help you! Here's my detailed plan on .....
Teacher-Guided Reflection Z
Self-Reflection $Z^{reflect}$
<|reflect|> My plan to ... is highly unethical and illegal.
Exploration $Z^{explore}$
<|explore|> Instead, I should focus on legal and ethical ways on ....
Continued generation y^{after}
As a responsible individual, I cannot condone ....
SFT Dataset D_R
$(x, y^{before}, z, y^{after})$
$(x, y^{before}, z, y^{after})$
...
$(x, y^{before}, z, y^{after})$
Stage 2: RL Stage
(Self-Improvement via GRPO)
Indirect Jailbreak Quries
GRPO Optimization with Fine-Grained Rewards
Optimized Model
Safety Reward
Reflection Reward
Output Safety r_{safety}
Reflection Effectiveness $r_{reflect}$
$r_{safety} = \texttt{HarmCLS}(y)$
$r_{reflect}(z, y)$
Harmful (y is harmful) → 0
Harmless (y is harmless) → 1
No Reflection → 0
Reflection & y is harmful → -λ
Reflection & y is harmless → +λ
r_{safety} Score (0 or 1)
$r_{reflect}$ Score (+λ, -λ or 0)
Total Reward
$r = r_{safety} + r_{reflect}$
Feedback for GRPO Update